

Combining qualitative and quantitative approaches for measuring underground economy of Pakistan

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Abstract The concern for underground, unofficial or shadow economy by researchers and practitioners has been enlarged in the past few decades. The existence and widespread findings of shadow economies is a growing problem around the world. In this study, we draw on insights from an active and growing economic literature on demand for currency approach and the underground economy. The study addresses the policy relevant issues in a technical fashion by using monetary approach with some extensions in the developing countries context like Pakistan over a period of 1975–2010. The study at first finds the long-run relationship between the currency in circulation and other variables (i.e., GNP, tax revenue as percentage of total revenue, interest rate on saving deposits and inflation rates) and subsequently used their results to infer the size of the informal sector in Pakistan's economy. The result shows that GNP and inflation rate both have a positive relationship with currency circulation, however, the magnitude of both variables on currency circulation is different, as if there is 1 % increase in GNP and price level, currency circulation increases by 0.86 and 0.091 % respectively. The other variables i.e., tax revenues and interest rates are negatively associated with the currency circulation in the long-run. However, these results are disappearing in the short-run. On the basis of cointegration results, the study concludes that Pakistan's underground economy increases from less than 30 % in 1975 to 19.8 % in 2010. A change in the size of the shadow economy reflected in a change in monetary indicators. Rising activity in the shadow economy is likely to push up the demand for currency in Pakistan.

Keywords Underground economy · Monetary approach · Cointegration · Pakistan

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Table 1 Types of underground economy

	Definition	Activities
Illegal economy	Totality of the revenues that are generated by those economic activities that violate the legal status of legitimate forms of trade	Illegal activities such as illegal drugs dealing, black market of currency exchange, money laundering, unlicensed money lending, illegal gambling prostitution and pornography
Unreported economy	Totality of the revenues that are generated by those economic activities that violate the legal status of legitimate forms of trade	Unreported revenue either from legal sources or illegal sources
Unrecorded economy	Activities that avoid institutional conventions that define the necessary requirements for the report to governmental agencies for statistics	–
Informal economy	Economic activities that avoid costs and excluded from the rights and benefits that come along with leasing, work contracts, loan and social security	Revenue that generated by economic agents that operate informally

Source [Feige \(1997\)](#) and [Radulescu \(2010\)](#)

1 Introduction

Economic activities can be classified under structural approach into two major group namely, formal economy and informal economy (underground economy). The underground economy can also be referred as parallel, irregular, cash, black, twilight subterranean, hidden, secondary, and shadow economy ([Kasipillai 1998](#)). Narrow underground economy may be described as legal, marketed transactions that are not included in measured gross domestic product (GDP), while fairly broader definition would include both legal and illegal marketed transactions that are not incorporated in GDP. The most comprehensive definition of underground economy would possibly be the one that integrated all transactions, legal or illegal, marketed or non-marketed that are either intentionally or unintentionally excluded from GDP ([Tedds 2005](#)). Underground economy can be of different forms and nature. It may be of illegal, unreported, unrecorded, and informal type. [Feige \(1997\)](#) reported different forms of underground economy which is shown in Table 1.

Lippert and Walker (1997) further suggest some classifications of underground economic activities in the form of monetary and non-monetary transaction, which are shown in Table 2.

1.1 Causes of underground economy

The main causes of underground economy are increase in tax burden, social security contributions, and enlarged regulation in the official economy particularly in labor markets, enforced drop of weekly working hours, earlier retirement and declining of tax morale ([Yilmaz and Ogunc 2000](#)). Taxes influence labor-leisure choices of economic agents and also encourage labor supply towards underground or untaxed part of economy. Higher tax and social security contributions results in lower after tax income for employers which can create an incentive for them to work in underground economy where they evade from taxes ([Startiené](#)

Table 2 Classification by types of underground economic activities

Activity category	Monetary transactions		Non-monetary transactions	
Illegal activities	Trade with stolen goods; production and dealing of drugs; prostitution; smuggling; gambling; fraud etc		Barter of drug, stolen goods, smuggling etc. Produce or growing drugs for own use. Theft for own use	
	Tax evasion	Tax avoidance	Tax evasion	Tax avoidance
Legal activities	Unreported income from self employment; wages, salaries, and assets from unreported work related to legal services and goods	Employee discounts fringe benefits	Barter of legal services and goods	All do-it-yourself work and neighbor help

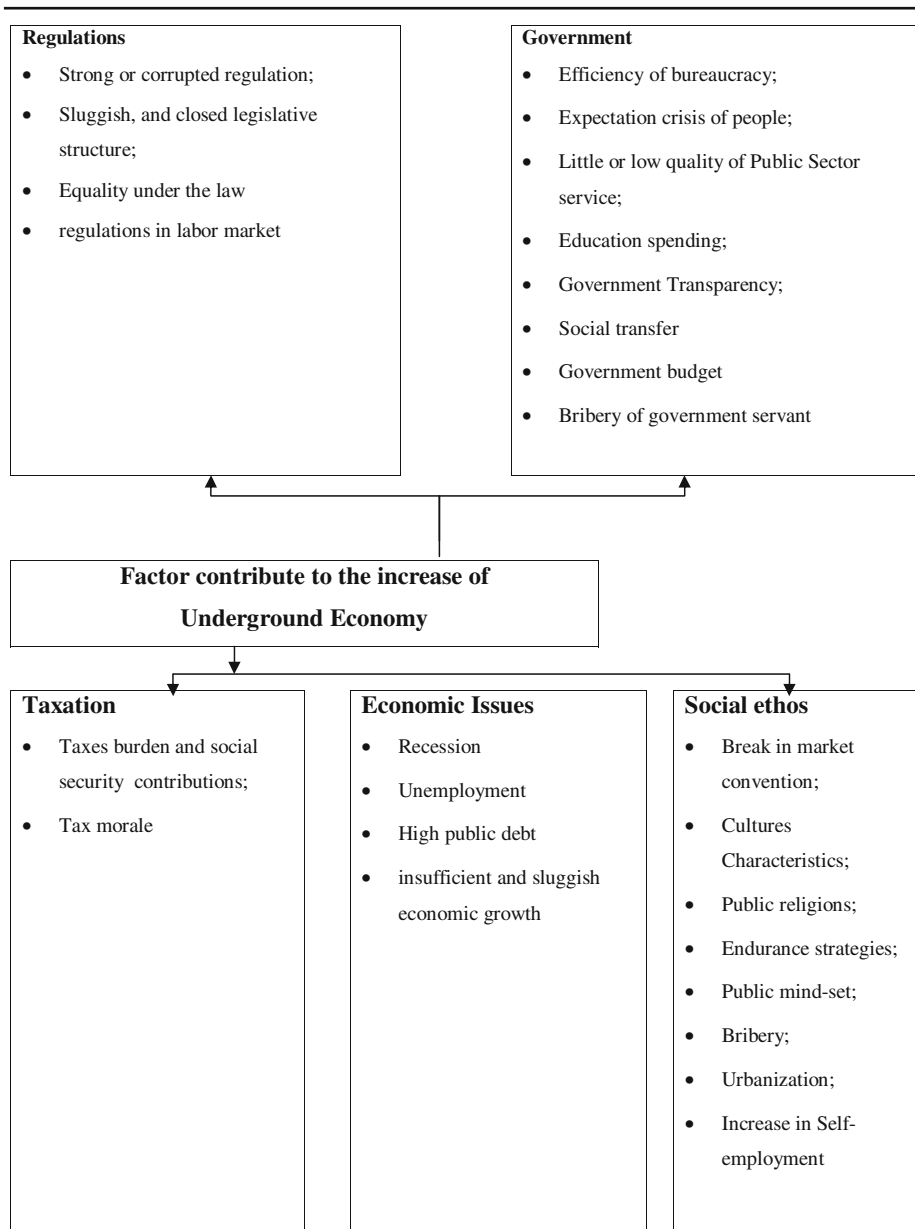
Source Lippert and Walker (1997)

and Trimonis 2010). The regulations can raise legal burden of employers and it is transferred onto employees in form of lower wage so it can drive employees to work in the underground economy (Schneider and Enste 2000). Enforced drop of weekly working hours and early retirement can cause employees to work on their potential working hours in underground economy (Startiené and Trimonis 2010). Public suspicions about government expenditures may decrease tax morale in a society. Public believe that tax revenue of state, which are paid by members of society, is spent inefficiently by government so it may prefer to pay tax as less as possible (Torgler et al. 2009).

The possibility and potential of having underground economy in urbanization is expected to be large when government cannot entirely satisfy public preferences and expectations (Torgler et al. 2009). Business and individuals use irregular activities to go underground as a mode of breaking in traditional markets to survive or even stay dominant (Marinov 2008). Various new generations from well-off families have begun to control their parents businesses, these new entrepreneurs are probably more involved in informal activities e.g. play trick in the law leakage, hide some of their business activities from the government, tax evasion, and avoid regular registration (Ott 1998). In economic recession, high unemployment rates, rising public debts, and more individual or enterprise bankruptcy cases, people try to compensate their financial losses through underground economy activities in order to survive in tough environment (Maurin et al. 2003; Schneider and Enste 2002). Table 3 summarizes diverse features which can contribute to aggregate size of underground economy.

1.2 Pakistan and underground economy

In case of Pakistan, underground economy grew in the 1960s and fell down to its lowest level in 1976, as underground economy is extremely sensitive to political and economic measures in country, it increased due to wars in 1965 and 1971 (Aslam 1998). Economic front of a country have positive effect and the share of underground economy reduces, it is true for mid seventies,

Table 3 Factors contribute to the increase of underground economy

Source Jie et al. (2011)

there was nearly no increase in tax rate which was a reason of underground economy to be under check in this period. Underground economy was increasing very swiftly from early 1980s due to afghan war in 1979, high tax regulations and introduction of structural reforms in 1989 (Kemal 2007). In 1990s, financial and banking sector reforms had dampening effect on

Table 4 Estimates of underground economy (% of total economy)

Time period	ARDL model	MIMIC	Electricity consumption approach
1960s	28.9	–	–
1970s	29.7	29.6	4.5
1980s	32.8	29.6	18.1
1990s	32.7	29.5	28.6
2000s	23.2	29.0	27.3

Source Arby et al. (2010)

underground economy because of higher involvement in formal economy. It declined in 1999–2002 when although economy grew with a reasonable average growth rate, underground economy declined is attributed to efficiency of tax authority for documentation, political stability and improved governance, but reverted to an increasing fashion until 2003. The destruction of 2005 Earthquake and calamities of recent flood in 2010 add a greater potential to the expansion of underground economy (Kemal 2007). Table 4 shows the underground economy as a percentage of total GDP of Pakistan during the last five decades.

1.3 Effects of underground economy

The growth of an underground economy can lead to considerable problems; firstly it creates trouble in determining status of activities in economy. It can also result in a lesser tax base and tax revenues, and might contribute to fiscal deficits (Kanao and Hamori 2010). Underground economy has negative as well as positive effects on formal or registered economy. On negative side, it creates an unfair competition circumstances for firms and out of blue worsen financial situation of social security institutions (Yilmaz and Ogunc 2000). Underground economy hampers fair and efficient resource allocation among different sectors of economy that enlarges tax burden on members of official economy (Yasmin and Rauf 2004). Stating positively about underground economy, it can have many fruitful effects on official economy through number of ways i.e., firms in underground economy may exhibit lower cost of production than registered firms. It will generate employment opportunities due to higher labor demand in formal economy which possibly reduce unemployment rate in overall economy. Moreover, society welfare level may improve as a result of underground economy (Yilmaz and Ogunc 2000).

It is thus vital to estimate unreported GDP in order to get the real picture of development in the economies. It has been a crucial point of study in Pakistan for a lot of years. Many researchers have made variety of efforts to quantify underground economy of Pakistan through numerous techniques. Continuous suffering of the country from low tax revenue collection and increasing budget deficit raised interest in this area of study. This study quantifies the size of underground economy which may contribute in policy making process and economic forecasting.

The objective of the study is to evaluate different methods for measuring the size of the underground economy. The more specific objectives are:

- i. To investigate the nature and existence of underground economy.
- ii. To estimate short-run and long-run relationship between currency circulations and other variables i.e., GNP, tax revenues, interest rate on saving deposits and inflation rate.
- iii. To estimate size of underground economy in Pakistan's by using currency demand approach (CDA).

The study divides in to the following sections, after introduction which is presented in Sect. 1 above; Sect. 2 shows the literature review. Data source and methodological framework are presented in Sect. 3. Results are discussed in Sect. 4. Final section concludes the study.

2 Literature review

There is a vast literature available on shadow, or underground, economy (Badaja 2002). While the definition of the shadow economy is subject to debate, a standard taxonomy attributes most of this activity to being the result of either (i) trade in illegal goods and services, or (ii) tax evasion. These activities are conducted in a way that leaves no evidence of the transaction, consequently, these types of transactions require ‘anonymity’ (e.g., lack of record keeping and communication of information) to avoid detection and punishment. As a result, official measures of GDP understate the value of economic activity (Gomis-Porqueras et al. 2011). Fisher’s (1979) used Fisher’s equation of exchange i.e., $MV = PT$, on the basis of demand deposits velocities, money circulation, and other exchange media to estimate size of US underground economy from 1939 to 1968. The result concludes that total size of underground economy was assumed to be best indicated by aggregate money supply. Tanzi (1980) estimated demand of currency equation and computed size of US underground economy. The result founds that changes in currency demand are outcome of underground economies. The estimated underground economy of US is between 2–7 % which shows the size of above ground economy. Shabsigh (1995) applied monetary approach to calculate underground economy of Pakistan from 1975–1990. Using ratio of currency in circulation to total demand deposits, the estimate size of the underground economy was 21 % of the total GDP in 1975 and decreased somewhat to 20.4 % in 1990, thus indicated a stagnant underground economy.

Ahmed and Ahmed (1995) adopted Tanzi’s monetary analysis to estimate size of underground economy in Pakistan for the period of 1960–1990. The addition of bearer bonds along with currency in circulation exposed that level of tax evasion has amplified over number of years but underground economy as a percentage of GDP showed a declined in Pakistan from 52 % in 1960 to 35 % in 1990. Loayza (1996) used an endogenous growth model whose production technology depends on congestible public services and their effects on informal sector of fourteen Latin American countries. The result founds that tax burden and labor market regulations increase underground economy activity, while power and efficiency of government institutions reduces underground economy. Aslam (1998) used Tanzi’s methodology to estimate magnitude of underground economy in Pakistan for the period of 1991–1998. The result concludes that the underground economy of Pakistan increase from 29 % in 1960 to 43.9 % in 1990. Doyle (1999) evaluated underground economy by using cointegration techniques to estimate currency demand functions for 15 industrialized nations and found that size of estimate of underground economy was less than zero.

Yilmaz and Ogunc 2000 applied indirect approaches consist of GDP approach, employment approach, tax auditing approach and monetary approach to Turkish data for measuring underground economy. The result concludes that lack of any well-established theory cause different results. Moreover, lack of necessary statistical data in many fields in Turkey, even if the required data was found, existence of differences between different sources also confines consistency of the results. Kyle and Warner (2001) estimated size of informal sector from five hundred thirty firms of Bulgaria. Results indicate that although size of underground economy has declined in mid 1990s, it remains a considerable percentage of Bulgarian economy. Improved tax collection with reduced taxes and deregulation affects its growth negatively.

Schneider (2003) used CDA and structural modeling approach to calculate size of underground economy in 110 developing, transitional and OECD countries. The average size of underground economy was estimated to be 41 % for developing countries, 38 % for the transitional countries and 18 % for the OECD countries, as a percentage of official GNI in year 2000. The key determinants of informal economy are high taxation and social security contributions combined with government regulations.

Bovi (2002) used panel regression techniques to analyze association between taxation, regulations, efficiency of bureaucracy and corruption in OECD countries. It was determined that underground economy was positively correlated largely with institutional failures and slightly with taxation and market regulations over 1990s. The result concludes that countries with largest informal economy have not essentially highest tax burdens, and countries with the top-level tax rate are not necessarily those with largest black economy. Nikoupour (2003) estimated size of underground economy in Iran from 1961–2001 by using modified currency ratio method. The result shows that mean size of underground economy to be 27.76 % of GDP during sample period. Savaşan (2003) estimated size of underground economy in Turkey by employing Multi-Indicators Multi-Causes (MIMIC) model and randomized response (RR) Technique over a period of 1970–1998. The result found that underground economy declined in the mid-1980s, due to structural changes in Turkish economy. The result concludes that all types of burden imposed by government are positively linked to the underground economy.

Kemal (2003) examine the size of informal economy for Pakistan from 1973 to 1902. The result found that underground economy of Pakistan enlarged from 20 % in 1974 to 54 % in 1998 and then declined to 37 % in 2002. Muller (2003) used three South African national household survey instruments i.e., 1993 Project Statistics for Statistics of Living Standards and Development; 1995, 1997 and 1999 October Household Surveys; and September 2000 Labor Force Survey to discover problems concerned in the informal sector. The study used resulting data sets to plan expansion in estimated size of informal sector. South Africa practiced increase in total recorded employment of about 20 % from 1993 to 2001, but most of this growth was attributed by informal sector and formal sector continues to contribute minute to job creation in South Africa. Dell'Anno (2003) estimated Italian underground economy by using structural equation approach and verified main criticisms about the consistency of "MIMIC method" (or model approach). Results confirmed a satisfactory reliability of this approach for estimating size of underground economy. Reyneri (2003) focused on the problems of high levels illegal immigration and a strongly established underground economy, faced by Southern European countries. The study concentrated on questions about who the immigrants are and what attracts them to the region and also provided a wide sketch of general employment situation in Southern Europe. The result found that problem of illegal immigration cannot be addressed without recognizing the complexities of their existing economic situation. Park and Hyun (2003) examine the determinants of tax compliance using experimental data for Korea. The result shows that tax payers have the same degree of compliance regardless of their income levels. Both the tax audit rate and the penalty rate are important deterrence from tax evasion, although the penalty rate is more effective. Tax payers have a strong tendency for a free-ride, which means less compliance for financing their government. Tax education is one of the effective tools to induce taxpayers to comply more.

Yasmin and Rauf (2004) adopted the monetary approach to measure underground economy through tax evasion in Pakistan over period 1974–1902. The results confirmed that underground economy has increased enormously from Rs. 12 billion in 1974 to Rs. 1085 billion in 2002. Tedds (2005) used MIMIC model for estimating hidden economy of Canada. The results indicate that underground economy in Canada grew steadily relative to measured GDP over the period of 1976–2001. Dreher and Schneider (2006) analyzed effect of shadow economy

on corruption for a cross section of 120 countries and a panel of 70 countries for period 1994–2002. The results show that shadow economy decrease corruption in high income countries, but enlarge corruption in low income countries. The study also found that heavy regulations increase both corruption and shadow economy. [Orviska et al. \(2006\)](#) estimates the relative size of shadow economy by using cross-section survey data based on individual responses to Slovakia and the Czech Republic in 2002. The result suggests that the relative figures of shadow economy for Slovakia and the Czech Republic are 23.2 and 21.8 % respectively.

[Bühn et al. \(2007\)](#) used multiple indicator multiple causes model (MIMIC) and of do-it-yourself (DIY) activities in Germany from 1970 to 2005. By 2005, the study found the level of shadow economy which has increased over the years. DIY activities have remained more or less constant since the early 1990s. Regulation and tax burden are motivating forces for the shadow economy while for DIY activities; the level of unemployment is the key factor. [Pau and Carolina \(2007\)](#) measured size of Netherlands shadow economy using two different estimation methods i.e. direct and indirect approach. Both methods gave similar results, showing size of underground economy of Netherlands Antilles to be around 11 % of GDP. [Kemal \(2007\)](#) revised their effort and used finest fit monetary approach to estimate underground economy and tax evasion for Pakistan from 1973 to 1995. The reorganized estimations indicated that underground economy and tax evasion were increasing swiftly in early 1980s and this rate accelerated in 1990s. The rate of increase slowed down till 1999 and then followed an increasing fashion till 2003. The underground economy ranges from 54.6 to 62.8 % of GDP in 2005 and the tax evasion ranges from 5.7 to 6.5 % of total GDP in 2005. [Bose et al. \(2008\)](#) examined impact of banking development on the size of shadow economies by using data on 119 countries for the period 1999 to 2005. The results show that an improvement in development of banking sector is linked with a smaller shadow economy in a broad cross-section of countries. Moreover strength and efficiency of banking sector can reduce size of a shadow economy.

[Kim \(2008\)](#) provides an economic model of tax evasion, which considers the interaction between tax authority and its sovereign government. It shows that tax evasion is influenced by the government's intention to control the economy, controlling for other determinants of tax evasion. [Vuletin \(2008\)](#) estimated size of underground economy for 32 Latin American and Caribbean countries in early 2000s. The study used a structural equation modeling approach and found that a rigid tax system and regulatory atmosphere, high inflation, and supremacy of agriculture sector are main factors in determining size of the informal economy. [Buehn and Schneider \(2009\)](#) used a structural equation model which extracts information on different dimensions of corruption and the shadow economy. The result shows a positive relationship between the shadow economy and corruption. [Hassan and Yu \(2010\)](#) examined size of underground economy in the U.S. between 1972 and 2006 by using dynamic error-correction-based CDA. The results show that income tax rates have no immediate effect on currency demand, but they have positive and considerable effects with a one-period lag. Size of underground economy in the U.S. has grown during the post-1990 period from \$148.7 billion in 1973 to \$691.2 billion in 2006. The study concludes that during recessionary periods, underground and official economies move in a same direction. [Arby et al. \(2010\)](#) estimated size of underground economy in Pakistan by using modified monetary approach by employing auto-regressive distributed lagged (ARDL) model based approach, electricity consumption approach and multi-indicators and multi-causes (MIMIC) model approach for the period 1966–1998. The results of modified monetary approach indicated that underground economy increased from 30 % in 1960s to 33 % in 1990s and then declined to 23 % in 2000s. The electricity consumption approach showed that informal economy amplified from about 5 % in 1970s to 29 % in 1990s and then declined to 27 % in 2000s. However, the MIMIC

model showed that underground economy was around 30 % of total GDP in Pakistan during study period.

[Kanao and Hamori \(2010\)](#) used dynamic ordinary least squares (DOLS) to estimate the size of underground economy of Japan. The results indicate that the size of the underground economy increased in the early 1990s but has been declining since the underground economy reached its climax in around 1992, approximating 25 % of nominal GDP. [Heinemann and Schneider \(2011\)](#) focused on religion's possible influence on shadow economy. Various dimensions of the religious markets were taken into consideration. The empirical test used largest available cross-section on size of shadow economy and matched this dataset with many religious indicators. The study found that indicators of religious competition do not have a quantifiable impact. Size of shadow economy is smaller in countries dominated by Islam or Eastern religions as compared to christian countries.

[Maria and Ștefănescu \(2011\)](#) estimated the size of underground economy in Romania by using the multiple indicators multiple causes (MIMIC) method. Results show that size of underground economy in Romania increased over time. Taxes, changes in official output per capita, unemployment and bureaucracy are main causes of Romanian underground economy. [Hassan \(2011\)](#) estimated the size of shadow economy on domestic sector, import sector and export sector in Bangladesh, using one survey and two model based techniques. The study found that size of shadow economy in Bangladesh is 10 % according to CDA, and 38 % according to MIMIC approach. Economic variables were also found to be more essential than policy variables in shadow economy sector. [Kholodilin and Thießen \(2011\)](#) investigated extent of shadow economy in 38 OECD countries by using panel-data techniques over sample period 1991–2007. The study found that estimates of shadow economy are extremely sensitive to the assumption on velocity of money circulation. It was also shown that micro- and macro-evidence are consistent at a fairly low velocity of money circulation.

[Gomis-Porqueras et al. \(2011\)](#) developed a dynamic, general equilibrium model of tax evasion. The shadow economy was measured by trade using cash to avoid taxes in their model. The study adjusted model using money, interest rate and GDP data to back out size of shadow economy for a sample of 30 countries and compared estimates to conventional estimates. The result shows that size of shadow economy is on average higher than those reported in the literature. [Singh et al. \(2012\)](#) analyzed different determinants of underground economy and focused on role of institutions and rule of law. The study found that businesses hide their activities in underground economy when they faced heavy regulation with inconsistent enforcement and corruption. It was empirically analyzed that institutions are more important determinant to size of underground economy than tax rates. [Gobbi and Zizza \(2012\)](#) investigate the relationship between underground activities and financial deepening in the Italian credit market. The results find a strong negative impact of the share of irregular employment on outstanding credit to the private sector i.e., a shift of 1 % of employees from regular activities to irregular ones corresponds to a decline of 1–2 % points in lending and of 0.3 % points in outstanding credit to households. [Blackburn et al. \(2012\)](#) examine the relationship between the underground economy and financial development in a model of tax evasion and bank intermediation. The result is that the marginal net benefit of income disclosure increases with the level of financial development which implies that the lower is the stage of such development, the higher is the incidence of tax evasion and the greater is the size of the underground economy. [Ardizzi et al. \(2013\)](#) assess the size of the underground economy by the traditional CDA in a panel of 91 Italian provinces for the years 2005–2008. The results show an average value of the shadow economy of 17.5 % of GDP. However, when the model does not account for the role played by criminal transactions, which amount, on average, to about 10 percent of GDP, the estimate of the underground economy increases up to 26.1 % of

GDP. Yfantopoulos (2013) examine the size and trends of the Greek underground economy over the period of 1958–2011. The results show that in the short run, 1 % point increase in the rate of economic growth reduce the underground economy by 0.13, and in the long run by 1.26.

From the above discussion, the present study found copious reason that ensure the existence of underground economy in various countries depending upon their macroeconomic policies, infrastructure and tax collection mechanism.

3 Data source and methodological framework

The data taken from World Bank Development Indicators, published by World Bank (WDI 2009), Annual reports of State Bank of Pakistan (SBP, various issues) and Economic survey of Pakistan (various issues). To estimate the informal economy, data over the annual frequencies from 1975–2010 was used on various economic, political, institutional and demographical variables.

3.1 Methods of estimating size of underground economy

3.1.1 Direct approaches

Direct approaches are called micro approaches as well, in view of the fact that they use well-designed surveys and samples based on voluntary replies or tax auditing. Surveys were used in many studies, but it has a big disadvantage. The reliability of survey results absolutely depends on respondents' answers. If respondent answers the questions without deception, survey can yield reliable conclusions. But if respondent does not answer the questions correctly, it yields misleading conclusions. This method can yield comprehensive information about underground economy when detailed questions are answered honestly. This is biggest advantage in favor of direct method. Underground economy can be estimated by comparing income declared for tax purpose and income measured by selective checks. Researcher aims to calculate unregistered economy by getting amount of undeclared task (Yilmaz and Ogunc 2000).

3.1.2 Indirect approaches

On contrary to micro approaches, indirect approaches are macroeconomic approaches. These approaches are also called indicator approaches as they make use of economic indicators that give information about expansion of underground economy over time. Indirect approaches consist of GDP approach, employment approach, tax auditing approach and monetary approach. Monetary approach consists of simple currency ratio method, transaction method and currency demand method (Yilmaz and Ogunc 2000).

GDP approach The ratio of difference between production measure of GDP and expenditure measure of GDP to production measure of GDP is used to obtain information about underground economy, during the application of *GDP approach*. This approach can give ambiguous conclusions if error part in income (or production) measurement process is high. Income measurement, especially expenditure measure of GDP may lead to high and unknown errors. Therefore, difference between these two aggregates is often recognized as omissions and error term. A further problem is seen in case of non disposable income that is earned from

underground economic activities. If income earned by underground economic activity is not spent, instead transferred overseas or stored in foreign currency, results of GDP approach will defer minimum level or rate for underground economy. So, consistent results could not be reached (Yilmaz and Ogunc 2000).

Employment approach It assumes rising underground economic activity (increasing employment in underground economy) when the ratio of employment to population is falling and the ratio of labor supply to population is being constant more or less. Firstly, this approach does not take in and measure second job owners. People can work in both formal and underground economy. But employment approach does not consider this issue. Secondly, changes in ratios may have other reasons for instance social reasons (migration from rural to urban areas or growing number of women in employment). Therefore, employment approach's indicators may be erratic and results of that approach may give misleading conclusions.

Tax auditing approach In *tax auditing approach*, tax authorities examine tax returns and determine undeclared income. That undeclared income amount is used for estimating underground economy. This approach has limited application in Pakistan due to limited tax audits.

Monetary approach As its name suggests; employs monetary statistics for estimating underground economy. This approach consists of simple currency ratio method, transaction method and currency demand method.

Simple currency ratio method According to *simple currency ratio method*, a climb in currency stocks and payments are sign of transactions, which are not registered by government.

$$Y_u = Y_r \frac{(C - K_r D)}{(K_r + 1) D}$$

This equation is mathematical illustration of the simple currency ratio method. According to this equation, any improvements in extent of official or registered economy will enlarge rather than lessen the estimated size of underground economy and anticipated ratio of underground economy to official economy is unchanged by improvements.

Transaction method In *transaction method*, when a certain period's official GNP value is subtracted from related period's GNP value that is determined by the related period's value of $M * V$, one can determine size of underground economy. To estimate size of underground economy, this approach requires determination of base year in which there is no underground economy. In the base year, ratio of $P * T$ to nominal GDP is assumed at normal level and it stays at its normal level, if underground economy does not exist. In spite of of theoretical strength of the transaction method, there are also numerous difficulties in application. Determining a base year with no underground economy is not an easily adequate assumption and assumption for constant normal ratio over time is also not up to standard. Moreover, one cannot obtain exact figures of the total volume of transactions. So there is also a problem of data availability in application stage. Due to shortcomings of above discussed methodologies, the present study used currency demand method for estimating size of underground economy in Pakistan.

Currency demand method The pioneering work in this area includes Guttman (1977), Feige (1979) and Tanzi (1983). Their approaches consist of specifying a demand for currency equation to be used to develop the effect of a change in the tax level on that demand. This method assumes that underground economic activities or transactions are realized only by using

cash or currency and underground economy's agents have a preference on cash payments in order to escape from government authorities. This basic opinion implies that an increase in underground economy will increase demand for money. Thus, in order to determine surplus demand part of the money demand, this method requires estimation of econometric currency demand equation in over time. Therefore, essence of this method is estimating currency demand equation. The second assumption of this method is linked to the velocity of money. Due to this assumption, the velocity of money in an official economy is equal to the velocity of money in underground economy. The third assumption explains fundamental reason for the existence of an underground economy. According to the third assumption, underground economy is caused by a tax burden, such as high tax rates. In this method, workers or people prefer to be in underground economy to escape from high tax burden.

Tanzi (1983) developed a currency demand equation i.e.,

$$\ln(C/M_2)_t = \beta_0 + \beta_1 \ln(1 + TW)_t + \beta_2 \ln(WS/Y)_t + \beta_3 \ln R_t + \beta_4 \ln(Y/N)_t + \mu_t, \quad (1)$$

where C/M_2 is the ratio of currency in circulation to broad money supply, TW is the weighted average tax rate, WS/Y is the proportion of wages and salaries in national income, R is the interest paid on saving deposits, Y/N is the per capita income

The present study modified this equation and derived the following model i.e.,

$$CC = \beta_0 + \beta_1 G + \beta_2 T + \beta_3 R + \beta_4 I + \mu_t, \quad (2)$$

where CC is the currency in circulation in real terms, G is the gross national product in real terms, T is the ratio of tax revenues to consolidated revenues, R is the 1 year nominal saving deposit interest rate and I is the rate of inflation

In Eq. (2), while currency in circulation in real terms is dependent variable, gross national product in real terms, ratio of tax revenues to consolidated revenues, 1 year nominal saving deposit interest rate and rate of inflation are used as independent variables.

GNP can cause an increase in circulating currency as expansion of the economy through rapid economic growth effect currency in circulation. Intensity of taxation causes people engage more and more in tax evasion that are facilitated by use of currency. Therefore, currency in circulation CC is expected to rise. The relationship between interest rate on nominal saving deposits and currency ratio is expected to be negative as a high interest rate on saving deposits may serve as an incentive for investment and public prefer to purchase saving deposits rather than holding currency (Keynesian monetary theory). Moreover, direct relationship between currency in circulation and inflation rate is expected as if price levels also double, causing inflation, amount of money in an economy needs to be double, because consumer pays twice as much for same amount of the good or service.

Application of conventional econometric methods for estimation of coefficients by using time series data is based on assumption that the model variables are stationary. A time series variable is stationary only if its mean value, variance and correlation coefficients remain constant through the time. If time series variables used in estimation of coefficients are non-stationary, then its R square coefficient may be of a high value and can cause an incorrect understanding about level of relation between variables although there may be no significant relation between variables.

In case of autoregressive distributive lags (ARDL), it is considered to obtain cointegration vector in mean squares method, not only the estimate of mean squares shall have normal distribution and also a smaller bias and greater efficiency shall be achieved with small samples (Pesaran and Shin 1995). The ARDL approach has the advantage that it does not require all variables to be $I(1)$ as the Johansen framework and it is still applicable if we have $I(0)$ and

I(1) variables in our set. As our variables in this model have mixed stationarity i.e. at levels and first difference, we used ARDL technique for estimation suggested by Pesaran and Shin (1999); Pesaran et al. (2001). The ARDL model allows the use of both stationary and non-stationary variables in one model and can also produce long run relationship. Pesaran and Shin (1999) showed that ARDL-based estimators are super-consistent, and valid inferences on the long-run parameters can be made. Likewise, they also find that suitable modification of orders of the ARDL model is adequate to simultaneously correct residual serial correlation and problem of endogenous regressors.

The present study set up a long run cointegration relationship between the currency in circulation and other related variables and then used it to infer the size of the informal sector in Pakistan's economy. The study assembled vector autoregression (VAR) of order p , denoted VAR (p), by following Pesaran et al. (2001), as:

$$Z_t = \mu + \sum_{i=1}^p \beta_i Z_{t-i} + \varepsilon_t, \quad (3)$$

where z_t is the vector of both x_t and y_t , where y_t is the dependent variable defined currency in circulation (CC), x_t is vector matrix which represents a set of explanatory variables i.e. gross national product in real terms, ratio of tax revenues to consolidated revenues, 1 year nominal saving deposit interest rate, rate of inflation, and t is a time or trend variable. According to Pesaran et al. (2001), y_t must be I(1) variable, but the regressors can be either I(0) or I(1). We further developed a vector error correction model (VECM) as follows:

$$\Delta Z_t = \mu + \alpha t + \lambda Z_{t-1} + \sum_{i=1}^{p-1} \gamma_i \Delta Y_{t-i} + \sum_{i=1}^{p-1} \gamma_i \Delta x_{t-i} + \varepsilon_t, \quad (4)$$

where Δ is the first-difference

The long-run multiplier matrix λ as:

$$\lambda = \begin{bmatrix} \lambda_{yy} & \lambda_{yx} \\ \lambda_{xy} & \lambda_{xx} \end{bmatrix}$$

The diagonal elements of the matrix are unrestricted, so the selected series can be either I(0) or I(1). If $\lambda_{yy} = 0$, then Y is I(1). In contrast, if $\lambda_{yy} < 0$, then Y is I(0).

The VECM procedures explained above are vital in the testing of at most one cointegrating vector between dependent variable y_t and a set of regressors x_t . To develop model, we followed the postulations made by Pesaran et al. (2001) in Case III, that is, unrestricted intercepts and no trends. After imposing the restrictions $\mu \neq 0$, $\lambda_{yy} = 0$ and $\alpha = 0$, the GIIE hypothesis function can be stated as the following unrestricted error correction model (UECM):

$$\begin{aligned} \Delta(CC)_t &= \beta_0 + \beta_1(CC)_{t-1} + \beta_2(G)_{t-1} + \beta_3(T)_{t-1} + \beta_4(R)_{t-1} + \beta_5(I)_{t-1} \\ &+ \sum_{i=1}^p \beta_6 \Delta(CC)_{t-i} + \sum_{i=0}^q \beta_7 \Delta(G)_{t-i} \\ &+ \sum_{i=0}^r \beta_8 \Delta(T)_{t-i} + \sum_{i=0}^s \beta_9 \Delta(R)_{t-i} + \sum_{i=0}^t \beta_{10} \Delta(I)_{t-i} + \mu_t, \end{aligned} \quad (5)$$

where Δ is the first-difference and μ_t is a white-noise disturbance term. CC is the currency in circulation in real terms, G is the gross national product in real terms, T is the ratio of tax revenues to consolidated revenues, R is the 1 year nominal saving deposit interest rate, I is the rate of inflation

Equation (5) also can be viewed as an ARDL of order (p, q, r, s , and t). This equation shows currency in circulation tends to be affected and explained by its precedent values. The structural lags are established by using minimum Akaike's information criteria (AIC). From the estimation of UECMs, the long-run elasticities are the coefficient of one lagged explanatory variable (multiplied by a negative sign) divided by the coefficient of one lagged dependent variable (Bardsen 1989). For example, in Eq. (5), the GNP, T, R and I are (β_2/β_1) , (β_3/β_1) , (β_4/β_1) and (β_5/β_1) respectively. The short-run effects are captured by the coefficients of the first-differenced variables.

After regression of Eq. (5), the Wald test (F -statistic) was used to differentiate long-run relationship between the variables. The Wald test can be carried out by imposing limitations on the estimated long-run coefficients of gross national product in real terms, ratio of tax revenues to consolidated revenues, 1 year nominal saving deposit interest rate, and rate of inflation variables. The null and alternative hypotheses are given as:

$$H_0 = \beta_1 = \beta_2 = \beta_3 = \beta_4 = \beta_5 = 0 \text{ (no long-run relationship)}$$

Against the alternative hypothesis:

$$H_1 = \beta_1 \neq \beta_2 = \beta_3 \neq \beta_4 \neq \beta_5 \neq 0 \text{ (a long-run relationship exists)}$$

The calculated F -statistic value will be evaluated with the critical values given in Table CI (iii) of Pesaran et al. (2001). According to these authors, the lower bound critical values assumed that the explanatory variables x_t are integrated of order zero, or $I(0)$, while the upper bound critical values assumed that x_t are integrated of order one, or $I(1)$. Therefore, if the computed F -statistic is lesser than lower bound value, then the null hypothesis is not rejected and we infer that there is no long-run relationship between currency in circulation and its determinants. On the other hand, if computed F -statistic is larger than the upper bound value, then currency in circulation and its determinants share a long-run level relationship. If the computed F -statistic falls between lower and upper bound values, in that case the results are indecisive.

Then the estimated parameters of Eq. (5), was used to calculate currency in circulation by using relevant variables. Then, currency in circulation estimated again by imposing zero tax rate value. The difference between these two estimates represents volume of circulating currency in underground economy. The nominal aggregates of underground economy were obtained by multiplying this difference with velocity of money.

4 Results and discussions

The present study first evaluated the time series data in terms of their being stationary or non-stationary so that we can approach a valid and reliable understanding about estimation of the model. The study used augmented Dicky–Fuller unit Root test and Phillips–Perron for checking stationarity of the variable. The results from augmented Dicky–Fuller unit root test are indicative of the fact that tax revenues/total revenues and inflation rate variables are stationary at level $I(0)$ as the absolute values of test statistic (τ) are higher than critical values at 1 % significance level. However, currency in circulation, GNP, and interest rate on saving deposits variables are non-stationary at level $I(0)$ as the absolute values of test statistic (τ) are less than critical values at 5 % significance level so we proceeded for first difference and all of them become stationary at first difference $I(1)$ the absolute values of test statistic (τ) are higher than critical values at 1 % significance level.

Table 5 ADF and PP unit root test

Variable	Level	First difference	Order of integration
Currency in circulation (CC)	-3.01	-4.13*	I(1)
GNP (G)	0.19	-6.35*	I(1)
Tax revenue/total revenues (T)	-7.13*	-3.26	I(0)
Interest rate on saving deposits (R)	-2.25	-6.55*	I(1)
Inflation rate (I)	-4.14*	-8.83	I(0)
Phillips–Perron test			
Currency in circulation (CC)	4.11	-4.12*	I(1)
GNP (G)	0.47	-6.35*	I(1)
Tax revenue/total revenues (T)	-6.28*	-3.25	I(0)
Interest rate on saving deposits (R)	-0.69	-6.54*	I(1)
Inflation rate (I)	-4.42*	-8.90	I(0)

*Indicate the rejection of null hypothesis of unit-root the series is non-stationary, or contains a unit root. The rejection of the null hypothesis is based on [Mackinnon \(1996\)](#) critical values at 1 % level of significance

Results of Phillips–Perron Test, shown in Table 5 are same as for augmented Dicky–Fuller unit Root test i.e. tax revenues/total revenues and inflation rate variable are stationary at level I(0) whereas currency in circulation, GNP and interest rate on saving deposits variables are not stationary at level rather they are stationary at first difference I(1).

The ARDL approach has the advantage that it does not require all variables to be I(1) as the Johansen framework and it is still applicable if we have I(0) and I(1) variables in our set. As our variables in this model have mixed stationarity i.e. at levels and first difference, we used ARDL technique for estimation suggested by [Pesaran and Shin \(1999\)](#) and [Pesaran et al. \(2001\)](#).

The estimation of Eq. (5) using the ARDL model is reported in Table 6. Using Hendry's general-to-specific method, the goodness of fit of the specification, that is, *R*-square and adjusted *R*-square, is 0.745 and 0.592 respectively.

The result shows that GNP and changes in price level (I) both have a significant and positive relationship with the currency circulation (CC), as 1 % increase in GNP and inflation, CC increases by 0.86 and 0.091 % respectively. While, total revenues (T) and interest rate (R) decreases CC by 0.092 and 0.075 % respectively in the long-run. The estimates of short-run coefficient reveal that interest rate (R) and GNP decreases CC by 0.083 and 0.718 % respectively, While, inflation rate increases CC by 0.078 % in the short-run.

In Table 7, results of the bounds co-integration test reveal that null hypothesis of against its alternative is rejected at 5 % significance level. The computed *F*-statistic of 4.873 is greater than the upper critical bound value of 4.01, thus indicating the existence of a steady-state long-run relationship among CC, GNP, T, R and I.

The estimated coefficients of long-run relationship between CC, GNP, T, R and I are expected to be significant, as:

$$D\log(CC)_t = -5.50 + 1.331\log(G)_t + 0.142\log(T)_t - 0.116\log(R)_t + 0.140\log(I)_t$$

$$(0.0688) (0.0004) (0.0496) (0.0073) (0.0040) \quad (6)$$

Note: bracket shows *p* values.

Table 6 ARDL estimations dependent variable: DLog (CC)

Variable	Coefficient	Std. error	t-Statistic	Prob
C	-5.509020	2.864254	-1.923370	0.0688
@TREND	0.033841	0.012489	2.709682	0.0135
Panel A: the long-run coefficient estimates				
LOG(CC(-1))	-0.649561	0.163015	-3.984659	0.0007
LOG(G(-1))	0.864977	0.290009	2.982583	0.0074
LOG(T(-1))	-0.092573	0.044304	-2.089484	0.0496
LOG(R(-1))	-0.075394	0.025261	-2.984573	0.0073
LOG(I(-1))	0.091044	0.028016	3.249764	0.0040
Panel B: the short-run coefficient estimates				
DLOG(R)	-0.083281	0.029597	-2.813833	0.0107
DLOG(I)	0.078972	0.022885	3.450815	0.0025
DLOG(G(-1))	-0.718430	0.348050	-2.064155	0.0522
DLOG(T(-1))	0.235169	0.132500	1.774861	0.0911
DLOG(I(-1))	-0.054138	0.022001	-2.460715	0.0231
DLOG(G(-2))	-0.743143	0.319660	-2.324792	0.0307

Statistical tests: $R^2 = 0.745$; Adjusted R -square = 0.592; Durbin Watson = 2.046, Wald test (F-statistics) = 4.873

Table 7 Bounds test for cointegration analysis

Critical value (%)	Lower bound value	Upper bound value
1	3.74	5.06
5	2.86	4.01
10	2.45	3.52

Wald F-statistics = 4.873. Critical values are cited from Pesaran et al. (2001), Table CI (iii), Case 111: unrestricted intercept and no trend

Taking into account the results obtained from Eq. (6) and Table 8 it is observed that the signs of all estimated coefficients are in line with theoretical principles and the coefficient is statistically significant. The variables have been applied in logarithmic form in the model, the estimated coefficients are indicative of elasticity concept. In other words, the percentage of variations in a dependent variable is indicative of variation in an independent variable for a variation unit. Hence, if there is one percent increase GNP, total taxes/total revenues and inflation rate, currency in circulation increases by 1.331, 0.142 and 0.140 % respectively, while interest rate on saving deposits is negatively related to currency in circulation i.e. if there is one percent increase in interest on saving deposits, currency in circulation decreases by 0.116 % this imply that GNP, total taxes/total revenues and inflation rate have a positive impact and interest rate have negative impact on currency in circulation which can cause money supply to increase and decrease respectively. Underground economy will increase with rise in money supply while decrease with fall in money supply (Yilmaz and Ogunc 2000).

The causality effect can be found by restricting the coefficient of the variables with its lags equal to zero (using Wald test). If the null hypothesis of no causality is rejected, then

Table 8 Long-run elasticities and short-run elasticities of currency in circulation in Pakistan: based on Eq. (6)

Long-run estimated coefficient variable	coefficient
G	1.331*
T	0.142**
R	−0.116***
I	0.140**
Short-run causality test (Wald test F-statistic)	
DLOG(R)	7.917 (0.010)*
DLOG(I)	11.908 (0.002)*
DLOG(G(-1))	4.260 (0.052)***
DLOG(T(-1))	3.150 (0.081)***
DLOG(I(-1))	6.055 (0.023)**
DLOG(G(-2))	5.404 (0.030)**

*, ** and *** Shows significance at 1, 5 and 10 % level

Table 9 Estimated short-run coefficients

Variable	Coefficient	Std. error	t-Statistic	Prob
DLOG(R)	−0.083281	0.029597	−2.813833	0.0107
DLOG(I)	0.078972	0.022885	3.450815	0.0025
DLOG(G(-1))	−0.718430	0.348050	−2.064155	0.0522
DLOG(T(-1))	0.235169	0.132500	1.774861	0.0911
DLOG(I(-1))	−0.054138	0.022001	−2.460715	0.0231
DLOG(G(-2))	−0.743143	0.319660	−2.324792	0.0307

we wrap up that a relevant variable effect currency in circulation. From this test, we initiate that all short run variables are statistically significant. To sum up the findings of the short-run causality test, we conclude that causality running from GNP, T, R and I Granger cause CC respectively. The results of short-run coefficients are shown in Table 9.

The results of Table 9 shows that GNP, total taxes/total revenues, interest rate on saving deposits, and inflation rate is significant in a short run, however, the intensity of impact on CC vary variable to variable. The foremost influence of GNP decreases CC by 0.718 %, followed by TR (i.e., 0.235 %), R (i.e., −0.083 %) and I (i.e., 0.078) percent respectively. Table 10 estimates the size of under ground economy by CDA for Pakistan.

The result shows that underground economy is increased with the decreasing rate. It was below 30 % during 1970s, increased to 32 % in 1990s and then declined to 21 % in current decade of 2000s. Not surprisingly, the underground economy was highest in the decade of 1990s i.e., a period of the most fragile and weak political regimes. It is only the decade of 2000s, when even though the economy grew with a reasonable average growth rate, the informality decreased attributable to activism by tax authority for documentation, political permanence and better governance.

The strength of the model has been definite by several diagnostic tests such as Breusch–Godfrey serial correlation LM test, ARCH test, Jacque–Bera normality test and Ramsey RESET specification test (Gujrati 2004). All the tests disclosed that the model has targeted econometric properties, it has a proper functional form and the model's residuals are serially uncorrelated, normally distributed and homoskedastic. Consequently, the outcomes reported are serially uncorrelated, normally distributed and homoskedastic. Hence, the results reported are valid for interpretation (see, Table 11).

Table 10 Estimates of underground economy (% of total economy)

Years	Underground economy (% of total economy)	Years	Underground economy (% of total economy)
1975	29.91032	1993	29.71801
1976	28.67874	1994	27.71912
1977	32.93365	1995	26.15022
1978	23.09819	1996	23.40507
1979	27.02552	1997	24.13617
1980	32.31107	1998	21.6791
1981	33.68443	1999	22.63289
1982	33.51186	2001	21.92331
1983	35.57455	2002	23.49339
1984	31.30486	2003	21.14723
1985	34.08069	2004	19.12673
1986	32.64783	2005	21.90056
1987	31.16641	2006	19.33415
1988	32.36394	2007	18.96763
1989	29.6068	2008	19.46562
1990	32.16616	2009	18.47447
1991	27.49589	2010	19.84525
1992	28.5224		

Table 11 Diagnostic tests

Tests	Chi-squares [<i>p</i> value]
$X^2_{Auto}(2)$	0.264 [0.8763]
X^2_{Norm}	3.625 [0.163]
X^2_{white}	0.437 [0.928]
X^2_{RESET}	0.012 [0.911]
$X^2_{ARCH}(1)$	0.259 [0.614]

The values in brackets are probabilities. $X^2_{Auto}(2)$ is the Breusch–Godfrey LM test for autocorrelation; X^2_{Norm} is the Jarque–Bera normality test; X^2_{white} is the White test for Heteroscedasticity; X^2_{RESET} is the Ramsey test for omitted variables and $X^2_{ARCH}(2)$ is the ARCH test for Heteroscedasticity

To check whether dependent variable i.e. growth rate (GNP) is stable or not over a course of time. We perform CUSUM and CUSUM square test which shows that growth rate is stable and significant at 5 % significant level. Figure 1 shows the CUSUM and CUSUM square stability test.

5 Summary and conclusion

Although there is neither an accurate definition of underground economy nor a technique exists that can measure it precisely, it is extremely vital to have some idea of its size because its

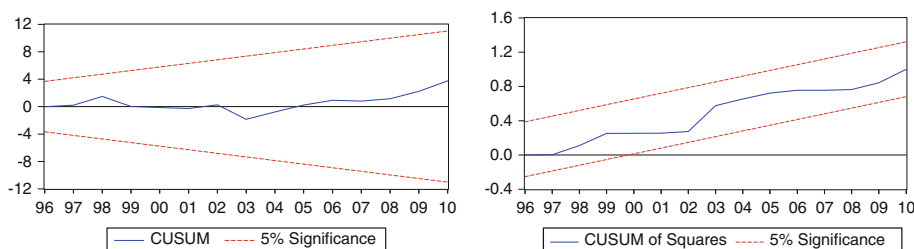


Fig. 1 CUSUM and CUSUM square test

existence renders the official values of macroeconomic indicators poor suggestion of real state of economy. Consequently, the economists have developed a number of direct and indirect methods to estimate the size the informal economy. The most commonly used method is monetary approach which assumes that underground transactions are taken place by using currency, the transaction velocity of money is equal in recorded and underground economies and that major purpose of being informal is tax avoidance. In a popular modification of this approach, a currency demand model is estimated with tax variable as one of the independent variable. The level of currency used in underground economy is found as a partial effect of tax variable. However, the way in which the earlier studies construct and estimate the model has several shortcomings.

This study addresses these shortcomings and uses autoregressive distributed lags (ARDL) method to estimate currency demand equation. We have used the estimated ARDL model to infer a long run relationship between currency demand and other variables including tax revenues to total revenues, GNP, inflation rate and interest rate on one year saving deposits. This relationship is then used to estimate underground economy in Pakistan. This study confirms the existence of a large underground economy in Pakistan over the period 1975–2010. The ARDL approach shows that the underground economy increased from less than 30 % in 1960s to 32 % in 1990s and then declined to 21 % on average in 2000s.

According to Kesselman (1997, p. 293),

There is little agreement about the size of the underground economy (UGE) relative to the total economy. No matter how small or how large the UGE might actually be, it has important implications for public policy.

The existence of the underground economy is a subject of severe concern for the government and policy makers. The presence of such a large underground economy is itself a sign of the existing corruption on the part of the public in Pakistan. This not only causes huge fiscal losses to economy but also causes inefficiencies in public administration. Report of the Auditor General of Canada to the House of Commons (Canada 1999) suggests some policy initiatives for underground economy which is shown in Fig. 2.

In Pakistan, there are a large number of economic activities which fall outside the reporting system and are beyond the reach of the tax net. Some of these activities are illegal, such as smuggling, narcotics or corruption. Other activities are legitimate but are not reported to tax authorities. Together, this group constitutes the informal, black or underground economy. The size of this economy has been growing faster than the formal economy (Iqbal and Qureshi 1998). To achieve the objective of inclusive growth and socio-economic development, the public policy may be devised with the sole objectives of increasing Tax to GDP ratio through expanding the tax base and plugging the tax leakages (Ahmed and Hussain 2006). The policy

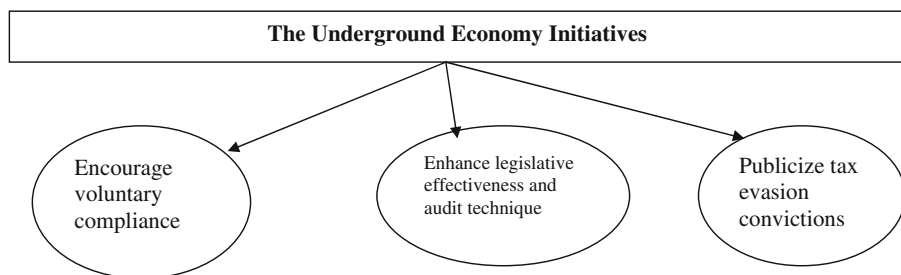


Fig. 2 Underground economy initiatives. Source: adapted from Canada (1999)

may also ensure a team of competent and honest people which may use those government funds in the most efficient and prudent manner to achieve the maximum social and economic welfare. These ultimate targets can be achieved through three intermediary targets i.e., first, formalizing the informal economy while retaining all its positive impacts and during this, searching for competent and honest people from the grass root levels; second, stopping the generation of informal income from within the formal sector (Arby et al. 2010); third, to stop the informal/improper implementation of rules and regulations within the formal sector (Gulzar et al. 2010).

Some of the suggestions arising from the results are given below:

- The Income tax unit must be well equipped with efficient staff required for suitable documentation.
- Tax collection system should be simple and comprehensive for the common people.
- A key objective of development plans should be concentrated on a campaign against underground economy and all preparative, executive and legal ways should be hired.
- The government should make a single provincial entity responsible for overseeing and coordinating a province-wide strategy to address the underground economy. This strategy could include links to federal and municipal initiatives.
- The Ministry of Labour, in collaboration with other regulators and levels of government, should take steps to acquire information and data that health and safety inspectors would use to identify and reduce underground economic activity.
- The Ministry of Labour should target workplaces and sectors operating in the underground economy for proactive inspections after normal working hours.

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